

Online

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 |
Generated: | Engine: DiscreteSim | Sections: 8

Offline and online

The same observations, refreshed periodically, with the decision they justify written down.

Source offline_cache status cached freshness unknown verdict keep (within_threshold)

The feed

Url https://raw.githubusercontent.com/caxelrud/Julia_Discrete_Sim_2026/main/data/online_feed.json
Policy offline_first
Status cached
Source offline_cache
Fetched at 2026-09-24T12:13:29
Freshness unknown
Attempts 2
Error TypeError: in keyword argument connect_timeout, expected Int64, got a value of type Float64
Fallback local_file

What arrived

Feed observations

Series	Observations
failure_interval	354
interarrival	4000
repair	354
service	4000
demand_size	82

Offline against online

Reevaluation

Parameter	Offline	Online	Relative change	Beyond threshold
arrival_rate	0.835	0.796	-0.047	no
service_rate	0.753	0.754	0.00238	no
mtbf	172.3	172.3	0	no
mttr	11.1	11.1	0	no
demand_mean	14.3	14.3	0	no

Are the two samples the same process?

Two sample ks

Series	Ks stat	Ks p	N offline	N online	Same process
failure_interval	0	1	354	354	yes
interarrival	0.019	0.125	54127	4000	yes
repair	0	1	354	354	yes
service	0.012	0.634	48780	4000	yes
demand_size	0	1	82	82	yes

The reevaluation log

Log

At	Source	Status	Freshness	Age days	Observations	Worst change	Verdict	Reason
2026-09-24T06:27:51	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T06:55:11	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T06:55:38	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold
2026-09-24T07:13:31	offline_cache	cached	unknown	—	8790	0.047	keep	within_threshold

The model always answers from local data first; the feed only ever improves the picture, so nothing here fails when the network is down.